Tutorial-09

Date:- 10/04/2024

A retail company wants to predict customer purchase behaviour using KNN and evaluate it using k-fold cross-validation.

Input features:

- Age
- Annual Income (in thousands)
- Shopping frequency (visits per month)
- Loyality Points

Target variable:

- purchase (yes=1, No=0).

Dataset:

CustomerID	Age	shopping Frequency	purchase
1	25	8	Y
2	45	3	N
3	28	9	Y
4	50	2	N
5	30	10	Y
6	48	3	N
7	26	8	Y
8	47	2	Y
9	29	9	N
			Y

3- Fold cross validation

cv fold = 3

KNN: K=3

9 . . . . 3 fold cv

$9/3 = 3$.

3 → Testing 6 → Training.

26,8)

$(25,8) = \sqrt{(26-25)^2 + (8-8)^2} = \sqrt{1^2+0^2} = 1.$

$(45,3) = \sqrt{(26-45)^2 + (8-3)^2} = \sqrt{(-19)^2 + 5^2} = \sqrt{361+25} = 19.6$

$(28,9) = \sqrt{(26-28)^2 + (8-9)^2} = \sqrt{(-2)^2 + (-1)^2} \sqrt{4+1} = \sqrt{5} = 2.23$

$(50,2) = \sqrt{(26-50)^2 + (8-2)^2} = \sqrt{(-24)^2 + 6^2} = \sqrt{576+36} = \sqrt{612} = 24.7$

$(30,10) = \sqrt{(26-30)^2 + (8-10)^2} = \sqrt{(-4)^2 + (-2)^2} = \sqrt{16+4} = \sqrt{20} = 4.47$

$(48,3) = \sqrt{(26-48)^2 + (8-3)^2} = \sqrt{(-22)^2 + 5^2} = \sqrt{484+25} = \sqrt{509} = 22.5$

Customer ID	Distance	Yes/No	
1	1	Yes	Yes
3	2.23	Yes	
5	4.47	Yes	

47,2)

$(25,8) = \sqrt{(47-25)^2 + (2-8)^2} = \sqrt{22^2 + (6)^2} = \sqrt{484 + 36} = \sqrt{520}$
$= 22.8$

$(45,3) = \sqrt{(47-45)^2 + (2-3)^2} = \sqrt{2^2 + (-1)^2} = \sqrt{4+1} = \sqrt{5} = 2.23$

$(28,9) = \sqrt{(47-28)^2+(2-9)^2} = \sqrt{19^2+(-7)^2} = \sqrt{361+49} = \sqrt{410} = 20.24$

$(50,2) = \sqrt{(47-50)^2+(2-2)^2} = \sqrt{(-3)^2+(0)^2} = \sqrt{9} = 3$

$(30,10) = \sqrt{(47-30)^2+(2-10)^2} = \sqrt{17^2+(-8)^2} = \sqrt{289+64} = \sqrt{353} = 18.78$

$(48,3) = \sqrt{(47-48)^2+(2-3)^2} = \sqrt{(-1)^2+(-1)^2} = \sqrt{1+1} = \sqrt{2} = 1.41$

Customer ID	Distance	yes/No.	
6	1.41	No	No
2	2.24	No	
4	3.	No	

(29,9)

$(25,8) = \sqrt{(29-25)^2+(9-8)^2} = \sqrt{4^2+1^2} = \sqrt{16+1} = \sqrt{17} = 4.12$

$(45,3) = \sqrt{(29-45)^2+(9-3)^2} = \sqrt{(-16)^2+(6)^2} = \sqrt{256+36} = \sqrt{292} = 17.0$

$(28,9) = \sqrt{(29-28)^2+(9-9)^2} = \sqrt{1^2+0^2} = \sqrt{1} = 1$

$(50,2) = \sqrt{(29-50)^2+(9-2)^2} = \sqrt{(-21)^2+(7)^2} = \sqrt{441+49} = \sqrt{490} = 22.13$

$(30,10) = \sqrt{(29-30)^2+(9-10)^2} = \sqrt{1^2+(-1)^2} = \sqrt{1+1} = \sqrt{2} = 1.41$

$(48,3) = \sqrt{(29-48)^2+(9-3)^2} = \sqrt{(-19)^2+(6)^2} = \sqrt{361+36} = \sqrt{397}$
$= 19.92$

Customer ID	Distance	yes/No	
3	1	yes	Yes
5	1.41	yes	
1	4.12	yes	

This is the 1st fold calculation. 10.04.16